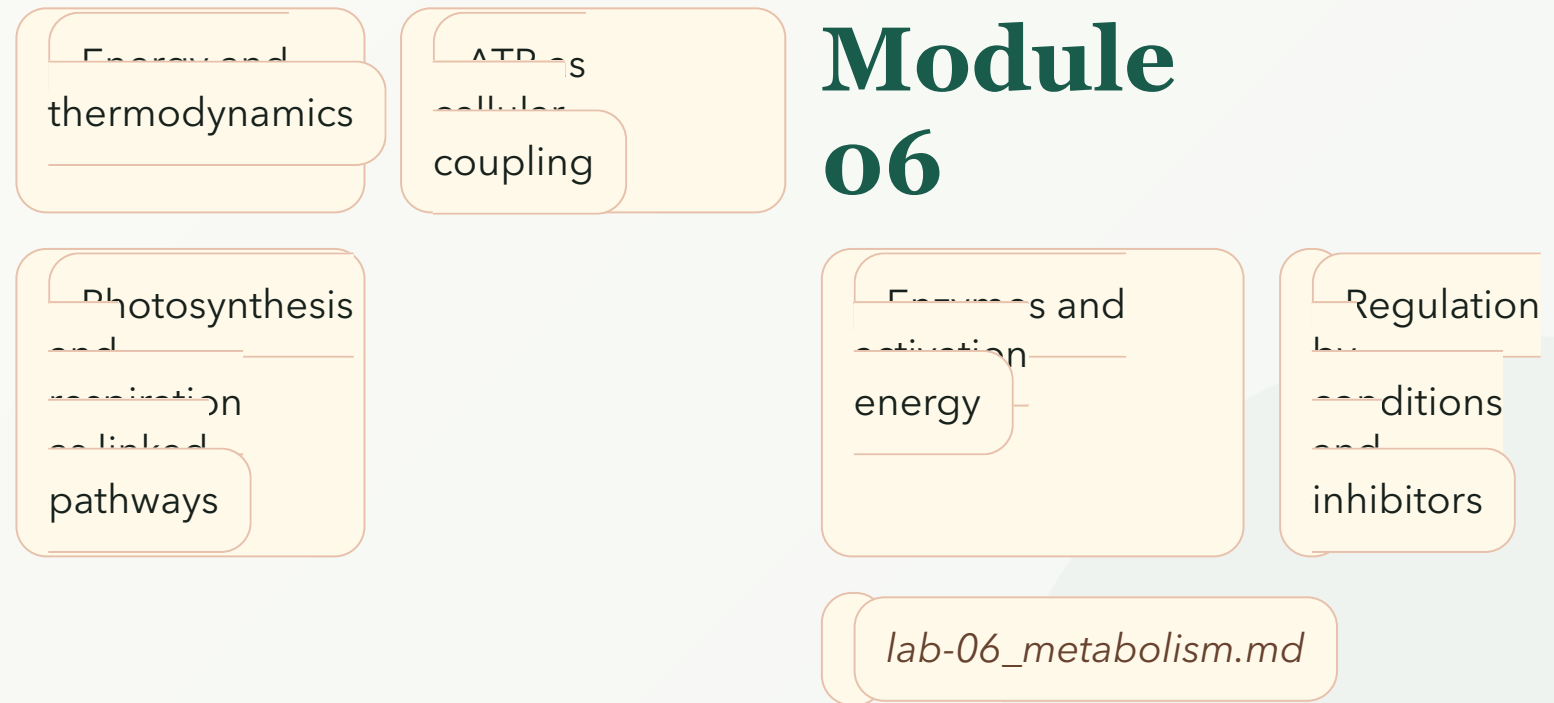


METABOLISM

Module map

- Module 06: Metabolism
- Connected lab:
lab-06_metabolism.md
- Use this deck as the visual path through the module's evidence and retrieval work.



METABOLISM

Learning objectives

- Apply thermodynamic ideas to living cells.
- Explain ATP as an energy-coupling molecule.
- Describe how enzymes change reaction rates.
- Predict how environmental conditions affect enzyme activity.
- Connect photosynthesis and cellular respiration in matter and energy flow.

OBJECTIVE 1

Apply thermodynamic ideas to living cells.

OBJECTIVE 2

Explain ATP as an energy-coupling molecule.

OBJECTIVE 3

Describe how enzymes change reaction rates.

OBJECTIVE 4

Predict how environmental conditions affect enzyme activity.

OBJECTIVE 5

Connect photosynthesis and cellular respiration in matter and energy flow.

METABOLISM

Topic sequence

- Energy and thermodynamics: Cells transform energy while obeying thermodynamic constraints.
- ATP as cellular coupling: ATP couples energy-releasing reactions to energy-requiring cellular work.
- Enzymes and activation energy: Enzymes lower activation energy without being consumed.
- Regulation by conditions and inhibitors: Temperature, pH, substrate concentration, and inhibitors alter enzyme function.
- Photosynthesis and respiration as linked pathways: Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

TOPIC 1

Energy and thermodynamics

TOPIC 2

ATP as cellular coupling

TOPIC 3

Enzymes and activation energy

TOPIC 4

Regulation by conditions and inhibitors

TOPIC 5

Photosynthesis and respiration as linked pathways

METABOLISM

Module 06: Metabolism Evidence Map

- Module focus: Metabolism
- Central claim: Metabolism explains energy transformations with ATP coupling, enzymes, and linked pathways.
- Key nodes to connect: Energy and thermodynamics, ATP as cellular coupling, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-06_metabolism.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

METABOLISM

Module 06: Metabolism Reasoning Sequence

- Module focus: Metabolism
- Trace the model: Energy and thermodynamics -> ATP as cellular coupling -> Enzymes and activation energy -> Regulation by conditions and inhibitors
- Inputs become outputs: Energy and thermodynamics, ATP as cellular coupling -> Apply thermodynamic ideas to living cells., Explain ATP as an energy-coupling molecule.
- Feedback check:
lab-06_metabolism.md evidence checks whether students can explain enzymes and activation energy.
- Lab connection:
lab-06_metabolism.md gives

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

METABOLISM

Terms as evidence handles

- Metabolism: All chemical reactions in a cell or organism.
- ATP: Cellular energy-coupling molecule.
- Activation energy: Energy required to start a reaction.
- Enzyme: Protein or RNA catalyst that speeds a reaction.
- Substrate: Reactant that binds an enzyme active site.
- Denaturation: Loss of shape that disrupts function.

METABOLISM

All chemical reactions in a cell or organism.

ATP

Cellular energy-coupling molecule.

ACTIVATION ENERGY

Energy required to start a reaction.

ENZYME

Protein or RNA catalyst that speeds a reaction.

SUBSTRATE

Reactant that binds an enzyme active site.

DENATURATION

Loss of shape that disrupts function.

METABOLISM

Lab connection

- Lab file: lab-06_metabolism.md
- Lab evidence should test or illustrate:
Enzymes and activation energy
- Students should leave with a
concrete observation, comparison, or
model tied to the module claim.

QUESTION

Energy and
thermodynamics

EVIDENCE

lab-06_metabolism.md

CLAIM

Apply thermodynamic
ideas to living cells.

METABOLISM

Contrast check

- Do not stop at: Cells transform energy while obeying thermodynamic constraints.
- Stronger explanation: Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Cells transform energy while obeying thermodynamic constraints.

DEEPER

Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

METABOLISM

Module 06: Metabolism Retrieval and Lab Check

- Module focus: Metabolism
- Retrieval prompt: How do the laws of thermodynamics constrain cells?
- Answer check: Apply thermodynamic ideas to living cells.
- Required terms: Metabolism, ATP, Activation energy
- Lab connection:
lab-06_metabolism.md;
lab-06_metabolism.md supplies evidence for metabolism evidence from reaction rate and condition changes.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

METABOLISM

Practice quiz bridge

- 1. Which statement best defines Metabolism in Module 06?
- 2. Which learning goal best supports the topic "ATP as cellular coupling"?
- 3. How should lab-06_metabolism.md support Module 06?
- 4. Which retrieval move best prepares a student for Metabolism?

QUIZ 1

A

QUIZ 2

B

QUIZ 3

C

QUIZ 4

D

METABOLISM

Synthesis and exit ticket

- Exit claim: explain energy and thermodynamics using evidence from lab-06_metabolism.md.
- Must include: Metabolism, ATP, Activation energy.
- Revision prompt: what evidence would change your explanation?

CLAIM

Energy and thermodynamics

EVIDENCE

lab-06_metabolism.md

REVISION

What would change your mind?